

Mechanical Engineering (Hinglish)

Heat Transfer

DPP: 1

Steady Conduction Heat Transfer

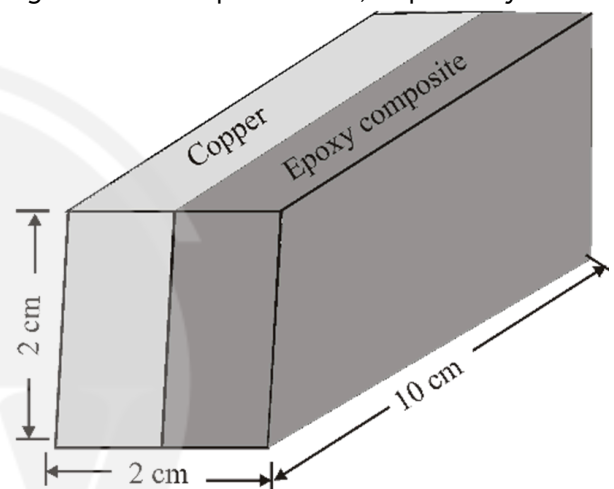
Q1 A hot fluid pipeline runs above the ground and are supported by a vertical steel shaft ($k = 25 \frac{\text{W}}{\text{mK}}$) that are long and have a cross sectional area of 0.005 m^2 . Under normal operating condition, the temperature variation along the length of a shaft is known to be governed by an expression of the form $T = 100 - 150x + 10x^2$, where T and x have unit of and meters respectively. Temperature variations are small over the shaft cross-section. Rate of heat loss from the side of the shaft ____ W (round off to one decimal place)

Q2 The temperature variation across a wall 0.3 m thick at a certain instant of time is $T(x) = a + bx + cx^2$ where T is in $^\circ\text{C}$ and x is in meter. $a = 200^\circ\text{C}$, $b = -200^\circ\text{C/m}$ and $c = 30^\circ\text{C/m}^2$. The wall has thermal conductivity of 1 W/mK . If the cold surface is exposed to a fluid at 100°C . Convective heat transfer coefficient (in $\text{W/m}^2 \text{ K}$) is ____ (round off to two decimal places).

Q3 The composite wall of an oven consists of three materials, two of which are of known thermal conductivity $k_A = 20 \text{ W/mK}$ and $k_C = 50 \text{ W/mK}$ and known thickness $L_A = 0.30 \text{ m}$ and $L_C = 0.15 \text{ m}$. The third material B, which is sandwiched between materials A & C, is of known thickness $L_B = 0.15 \text{ m}$, but unknown thermal conductivity k_B . The air is at 800°C and convective heat transfer coefficient between air and wall A is $25 \text{ W/m}^2\text{K}$. If outer surface of A and B are at 600°C and 20°C respectively then, under steady conditions k_B is ____ W/mK (round off to two decimal places)

Q4 A 10 cm long bar with a square cross section is shown in figure. The bar consists of a 1 cm thick

copper layer ($k = 400 \text{ W/m-K}$) and a 1 cm thick epoxy composite layer ($k = 0.4 \text{ W/m-K}$). What will be the rate of heat transfer under a thermal driving force of 50°C , when the direction of steady one-dimensional heat transfer is from front to back (i.e. along its length), from left to right and from top to bottom, respectively?



- (A) 40 W , 4 W , 1001 W
 (B) 4 W , 40 W , 1001 W
 (C) 40 W , 1001 W , 4 W
 (D) 1001 W , 4 W , 40 W

Q5 The refrigerator wall is constructed of two layers of 1 mm thick sheet metal ($k = 15.1 \text{ W/m} \cdot ^\circ\text{C}$) and fiberglass insulation ($k = 0.035 \text{ W/m} \cdot ^\circ\text{C}$) is used between these two layers to avoid heat transfer. The average heat transfer coefficients at the inner and outer surfaces of the wall are $4 \text{ W/(m}^2 \cdot ^\circ\text{C)}$ and $9 \text{ W/(m}^2 \cdot ^\circ\text{C)}$, respectively. The refrigerated space is maintained at 3°C and the average kitchen temperature is 25°C . It is observed that condensation occurs on the outer surfaces of the refrigerator when the temperature of the outer surface drops to 20°C . What will be the minimum thickness of fiberglass insulation that needs to be used in the wall in



order to avoid condensation on the outer surfaces?

- (A) 67 cm
- (B) 0.45 cm
- (C) 50 cm
- (D) 0.38 cm

Q6 Given: The composite wall having unit length normal to the plane of paper is shown in the figure.

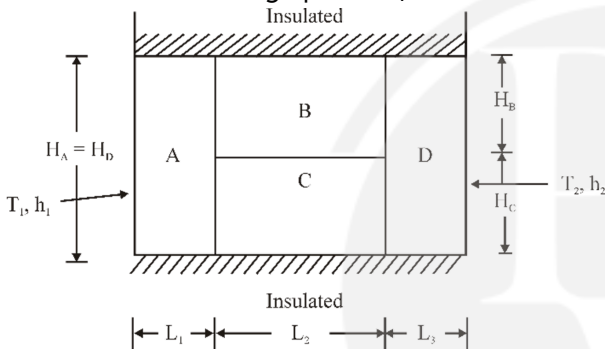
$$H_A = H_D = 3 \text{ m}, H_B = H_C = 1.5 \text{ m}$$

$$L_1 = L_3 = 0.05 \text{ m}, L_2 = 0.1 \text{ m}$$

$$k_A = k_D = 50 \text{ W/mK}, k_B = 10 \text{ W/mK}, k_C = 1 \text{ W/mK}$$

$$T_1 = 200^\circ\text{C}, h_1 = 50 \text{ W/(m}^2 \text{ K)}, T_2 = 25^\circ\text{C}, h_2 = 10 \text{ W/(m}^2 \text{ K)}.$$

Which of the following options is/are correct?



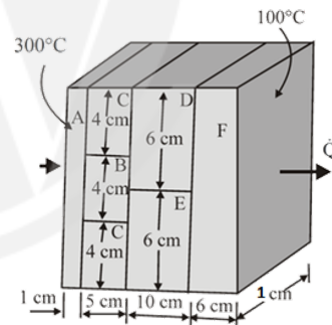
- (A) The rate of heat transfer through the wall lies between 3.70 to 3.80 kW/m.
- (B) The equivalent thermal resistance lies between 0.080 to 0.090 mK/W.
- (C) The rate of heat transfer through the wall lie between 5.50 to 5.60 kW/m.
- (D) The equivalent thermal resistance lies between 0.040 to 0.050 mK/W.

Q7 A square plate 1 kW electric heater (150 mm × 150 mm) is inserted between two slabs. Slab A is

20 mm thick ($k = 50 \text{ W/mK}$) and slab B is 10 mm thick ($k = 0.2 \text{ W/mK}$). The outside heat transfer coefficients on side A and side B are $200 \text{ W/(m}^2 \text{ K)}$ and $50 \text{ W/(m}^2 \text{ K)}$ respectively. The temperature of surrounding air is 25°C .

- (A) the maximum temperature in the system lies between 245 to 250°C .
- (B) the outside surface temperatures of the slab A lies between 230 to 233°C .
- (C) the outside surface temperatures of the slab B lies between 87 to 90°C .
- (D) the maximum temperature in the system lies between 180 to 185°C .

Q8 For a 0.22 m thick wall as shown in the figure the thermal conductivities of various materials used in $\text{W/(m}\cdot^\circ\text{C)}$ are $k_A = k_F = 2$, $k_B = 8$, $k_C = 20$, $k_D = 15$ and $k_E = 35$. The left and right surfaces of the wall are maintained at uniform temperatures of 300°C and 100°C , respectively. Disregard any contact resistances at the interfaces. If heat transfer through the wall is to be one dimensional, what will be the temperature at the point where the sections B, D and E meet?



- (A) 225°C
- (B) 197°C
- (C) 263°C
- (D) 288°C



Answer Key

Q1 2.3~2.7

Q2 4~4.44

Q3 1.4~1.6

Q4 (A)

Q5 (B)

Q6 (A, D)

Q7 (A, B, C)

Q8 (C)



[Android App](#)



[iOS App](#)



[PW Website](#)



ME & XE

Heat Transfer

DPP 01

Conduction Heat Transfer

- Q1** The inner & outer surface of a furnace wall, 40 cm thick, are at 300°C & 50°C respectively. The thermal conductivity of the wall material varies with temperature & is given by $k = 0.005 T - 5 \times 10^{-6} T^2$, where T is temperature in $^{\circ}\text{C}$. Heat loss per square meter surface area is (in W/m^2)
- (A) 320 (B) 435
(C) 490 (D) 560
- Q2** The temperature distribution across a large concrete slab 50 cm thick heated from one side is measured by thermocouples approximates to the following relation
- $$T = 60 - 50x + 12x^2 + 20x^3 - 15x^4$$
- Where T is in $^{\circ}\text{C}$ and x is in meter. Considering an area of 5 m^2 , location where the rate of heating or cooling is maximum is (in m)_____.
- Q3** The plane wall A of thickness 5 cm & thermal conductivity $80 \text{ W}/(\text{m}^{\circ}\text{C})$ & having volumetric heat generation rate of $1.25 \text{ MW}/(\text{m}^3)$ is insulated on one side while the other side is in contact with surface of another wall B. The wall B has no heat generation, is made of material of thermal conductivity $160 \text{ W}/(\text{m}^{\circ}\text{C})$ & has thickness of 2.5 cm. The non-contact surface of the wall B is exposed to the cooling fluid at 25°C . Convective heat transfer coefficient between wall B & the fluid is $1000 \text{ W}/(\text{m}^2\text{K})$. Maximum temperature (in $^{\circ}\text{C}$) is _____.
- Q4** A plane slab of 100 mm thickness generates heat. It is observed that the temperature drops between the centre and its surface to be 50°C . If the thickness is increased to 20 cm, then the temperature difference will be (Assume symmetric condition)
- (A) 100°C (B) 200°C
(C) 400°C (D) 600°C
- Q5** Consider steady state one dimensional heat conduction through a plate of 30 mm thickness with a uniform heat generation of $80 \text{ MW}/\text{m}^3$. The left and right faces are kept at constant temperature of 180°C and 120°C respectively. The plate has constant thermal conductivity of $200 \text{ W}/\text{m-K}$. Choose the correct statements.
- (A) The direction of heat transfer will be from the surface at 180°C to the surface at 120°C only.
(B) The temperature distribution is unsymmetric about the mid plane of the wall.
(C) Location of maximum temperature within the plate from its left face is 10 mm.
(D) The maximum temperature within the plate is 200°C .
- Q6** A conical section fabricated from pure aluminium. It is of circular cross section having diameter $D = ax^{\frac{1}{2}}$ where x is in m and $a = 0.5 \text{ m}^{1/2}$. The small end is located at $x_1 = 25 \text{ mm}$ and the large end at $x_2 = 125 \text{ mm}$. The end temperatures are $T_1 = 600 \text{ K}$ and $T_2 = 400 \text{ K}$, while the lateral surfaces are well insulated. The rate of heat flow (in kW) is _____. Take k for Al = $236 \text{ W}/\text{mK}$
- Q7** An 8 cm thick plate wall generates heat at the rate of $1.2 \times 10^5 \text{ W}/\text{m}^3$. One side of the wall is



exposed to environment at 90°C while the other side is insulated. The convective heat transfer coefficient between the wall and environment is $560 \text{ W}/(\text{m}^2\text{C})$. Thermal conductivity of the wall material may be taken as $0.15 \text{ W}/\text{m}^{\circ}\text{C}$. Maximum temperature to which the wall will be subjected is _____ (in $^{\circ}\text{C}$).

Q8

A wall of 0.5 m thickness is to be constructed from a material which has an average thermal conductivity of $1.4 \text{ W}/\text{m-K}$. The wall is to be insulated with a material having an average thermal conductivity of $0.35 \text{ W}/\text{m-K}$, so that the heat loss per square meter will not exceed 145 watt. Inner and out surfaces are 120°C and 15°C respectively. Thickness of insulation required is _____m.



Answer Key

Q1 (B)

Q2 0.3~0.36

Q3 115~118

Q4 (B)

Q5 (B, C, D)

Q6 5.65~5.85

Q7 2600~2700

Q8 0.12~0.135



[Android App](#)

|

[iOS App](#)

|

[PW Website](#)



ME & XE

Heat Transfer

DPP: 1

Critical Radius of Insulation

- Q1** With an increase in the thickness of insulation around a circular pipe, heat loss to surroundings due to
 (A) convection increases while heat due to conduction decreases
 (B) convection decreases, while that due to conduction increases
 (C) convection and conduction decreases
 (D) convection and conduction increases
- Q2** If the radius of a current carrying wire is less than the critical radius, then the addition of electrical insulation will enable the wire to carry a higher current because
 (A) the thermal resistance of the insulation is reduced.
 (B) the thermal resistance of the insulation is increased.
 (C) the heat loss from the wire would decrease.
 (D) the heat loss from the wire would increase.
- Q3** A sphere of 5 mm diameter at 50°C , is covered by a 1 mm thick plastic insulation. Thermal conductivity of sphere material and insulation material is 20 W/mK and 0.13 W/mK respectively. The sphere is exposed to a medium at 15°C , with a combined convection and radiation heat transfer coefficient of $20 \text{ W/m}^2\text{-}^{\circ}\text{C}$. Providing insulation on the sphere,
 (A) Always increase the heat transfer
 (B) Always decrease the heat transfer
 (C) Increase the heat transfer up to critical radius then decrease.
 (D) Decrease heat transfer up to critical radius and then increases.
- Q4** It is desired to increase the heat dissipation rate from the surface of an electronic device of spherical shape of 5 mm radius exposed to convection $h = 10 \text{ W/m}^2\text{K}$ by encasing it in a spherical sheath of conductivity $k = 0.04 \text{ W/mK}$. For maximum heat flow, the thickness of sheath should be,
 (A) 3 mm (B) 8 mm
 (C) 5 mm (D) 10 mm
- Q5** For pipes, the radius of the pipe is taken higher than the critical radius, so that any insulation added will only
 (A) decrease the heat loss from the pipe.
 (B) increase the heat loss from the pipe.
 (C) keep the heat loss unaltered.
 (D) enable heat gain from the surroundings.
- Q6** A 2 mm diameter wire with 0.8 mm thick layer of insulation ($k = 0.15 \text{ W/m-K}$) is used in a certain electric heating application. The insulated surface is exposed to atmosphere with convective heat transfer coefficient $40 \text{ W/m}^2\text{-K}$. The percentage change in heat transfer rate would occur if critical thickness of insulation is used, is _____% (round off to one decimal place). It may be assumed that temperature difference between surface of the wire and surrounding air remains unchanged.
- Q7** A wire of radius 3 mm and 1.25 m length is to be maintained at 60°C by insulating it by a material of thermal conductivity 0.175 W/mK . The temperature of surrounding air is 20°C with heat transfer coefficient $8.5 \text{ W/m}^2 \text{ K}$. For maximum heat dissipation, which of the following options is/are correct?
 (A) The minimum thickness of insulation is in the range of 17 to 18 mm.
 (B)



The rate of heat loss is in the range of 18.50 to 19.00 Watt.

- (C) The percentage increase in heat loss due to insulation is in the range of 133 to 136%.
- (D) The percentage increase in heat loss due to insulation is in the range of 83 to 86%.

Q8 A copper tube of 20 mm outer diameter, 1 mm thickness and 20 m long (Thermal conductivity = 400 W/m-K) is carrying saturated steam at 150°C (Convective heat transfer coefficient =

$150 \text{ W/m}^2\text{-K}$). The tube is exposed to an ambient temperature of 27°C . The convective heat transfer coefficient of air is $5 \text{ W/m}^2\text{-K}$. Glass wool is used for insulation (Thermal conductivity = 0.075 W/m-K). If the thickness of the insulation used is 5 mm higher than the critical thickness of insulation, the rate of steam condensation in kg/hr is _____ (correct up to two decimal places).

The enthalpy of condensation of steam = 2230 kJ/kg



Answer Key

Q1 (A)

Q2 (A, D)

Q3 (C)

Q4 (A)

Q5 (A)

Q6 14.5~15.5

Q7 (A, B, C)

Q8 1.2~1.3



[Android App](#) | [iOS App](#) | [PW Website](#)



ME & XE (Hinglish)

Heat Transfer

DPP: 1

Fins

Q1 Choose the correct statement corresponding to fins.

- (A) Fins are equally effective irrespective of whether they are on the gas side or liquid side.
- (B) For $\frac{h}{mk} > 1$, adding an extended surface increase the rate of heat transfer.
- (C) For $\frac{h}{mk} > 1$, adding an extended surface decrease the rate of heat transfer.
- (D) The fin effectiveness is enhanced by using a material of high thermal conductivity.

Q2 A 1 cm diameter, 30 cm long fin made of aluminium ($k = 237 \text{ W/m}^\circ\text{C}$) is attached to a surface at 80°C . The surface is exposed to ambient air at 22°C with a heat transfer coefficient of $11 \text{ W/m}^2 \text{ }^\circ\text{C}$. If the fin can be assumed to be very long, the efficiency is _____% (round off to nearest integer).

Q3 A very long rectangular fin of 5 cm width, 1 mm thickness, 200 W/m-K thermal conductivity and 40°C base temperature is attached to the flat surface such that the temperature at the end of the fin is same as that of the surrounding air that is 20°C . Heat transfer coefficient is $20 \text{ W/m}^2\text{-K}$. Which of the following options is/are correct?

- (A) Fin temperature at a distance of 5 cm from the base is in the range of 29.5°C to 30.5°C .
- (B) Fin temperature at a distance of 5 cm from the base 34.5°C to 35.5°C .
- (C) The rate of heat loss from the entire fin is in the range of 2.7 to 3.2 W.
- (D) The rate of heat loss from the entire fin is in the range of 5.3 to 5.7 W.

Q4

An electronic device generates 0.16 kJ/hr of heat. To keep the surface temperature at the upper safe limit of 75°C , it is desired that the generated heat should be dissipated to the environment which is at 30°C . For this square aluminum fins of size 0.5 mm and length 10 mm are attached to the surface. Thermal conductivity of fin material is $690 \text{ kJ/m-hr-}^\circ\text{C}$ and heat transfer coefficient is $45 \text{ kJ/m}^2\text{-hr-}^\circ\text{C}$. Neglect the heat loss from the fin tip. Number of Fins required is _____ (round off to nearest integer).

Q5 A long copper rod ($k = 400 \text{ W/mK}$) 20 mm in diameter extends horizontally from a plane heated wall maintained at 100°C . The surface of the rod exposed to air environment at 20°C . Temperature variation along the rod is given by $T(x) = T_a + (T_b - T_a)e^{(-2x)}$ where x is the distance measured from wall (in m). Rate of heat transfer through the fin is (in watt) _____ (round off to one decimal place).

Q6 One end of two long cylindrical rods A and B of the same diameter but different materials, is attached to a base surface maintained at 120°C . The surface of the rods is exposed to ambient air at 30°C . By traversing the length of each rod with thermocouple, it was observed that the temperature of the rods is equal at the distance $x_A = 0.18 \text{ m}$ (for rod A) & $x_B = 0.078 \text{ m}$ (for rod B), where x is measured from the base. If the thermal conductivity of rod A is 70 W/m-K , the thermal conductivity of rod B will be (in W/m-K) _____ (round off to two decimal places).

Q7



Two long rods of the same diameter, one made of the brass ($k = 85 \text{ W/m-K}$) and the other of copper ($k = 375 \text{ W/m-K}$), have one of their ends inserted into a furnace. At a section 10.5 cm away from the furnace, the temperature of the brass rod is 120°C . Distance from the furnace end, the same temperature would be reached in the copper rod (in cm) is _____ (round off to two decimal places).

Assume both the rods are exposed to the same environment.

- Q8** A 50 mm × 50 mm iron bar 0.5 m long is connected to the wall of two heated reservoir, each at 100°C . The ambient air temperature is 30°C and convective heat transfer coefficient is $17 \text{ W/m}^2\text{-K}$. Thermal conductivity of iron bar is 50 W/m-K . Temperature of the bar midway between the reservoir (in $^\circ\text{C}$) is _____ (round off to one decimal place).



Answer Key

Q1 (C, D)

Q2 76~78

Q3 (A, C)

Q4 4~4

Q5 19~21

Q6 12.5~13.5

Q7 21~23

Q8 65~66



[Android App](#) | [iOS App](#) | [PW Website](#)



ME & XE (Hinglish)

Heat Transfer

DPP: 1

Unsteady/Transient Conduction Heat Transfer

- Q1** The lumped parameter analysis method for transient heat conduction problems is used if,
 (A) Physical size of the body is very small.
 (B) Thermal conductivity of the material is very large.
 (C) Convective heat transfer coefficient is very small.
 (D) Temperature gradient with the solid is very small.
- Q2** Heisler Charts shows the temperature time history of a solid in transient heat conduction as a function of
 (A) Fourier Number and Biot number
 (B) Fourier number and reciprocal of Biot number
 (C) Reciprocal of Fourier number and Biot number
 (D) Reciprocal of Fourier number and reciprocal of Biot number
- Q3** A copper thermocouple, initially at 25°C , which when placed in a gas stream at 200°C measures a temperature of 198°C in 5 seconds. For copper, $\rho = 8940 \text{ kg/m}^3$, $c = 384 \text{ J/kg-K}$, $k = 390 \text{ W/m-K}$ and the convective heat transfer coefficient $= 400 \text{ W/m}^2\text{K}$. The diameter of the thermocouple is ____mm (round off to two decimal places).
- Q4** A cylindrical steel ingot (diameter 100 mm, length 300 mm, $k = 40 \text{ W/mK}$, $\rho = 7600 \text{ kg/m}^3$ and $c = 600 \text{ J/kg-K}$) is to be heated in a furnace from 50°C to 850°C . The temperature inside the furnace is 1300°C and the surface heat transfer coefficient is $100 \text{ W/m}^2\text{-K}$. The time required for heating in minutes is ____ (round off to one decimal place).
- Q5** 900°C is placed in air at 30°C . For steel, $\rho = 7800 \text{ kg/m}^3$, $c = 460 \text{ J/kg }^{\circ}\text{C}$ and $k = 40 \text{ W/mK}$ and the convective heat transfer coefficient is $20 \text{ W/m}^2\text{-K}$. Rate of cooling after 30 seconds (in $^{\circ}\text{C/min}$) is
 (A) 17.3 (B) 20.8
 (C) 25.6 (D) 45.5
- Q6** Choose the correct statement.
 (A) Lumped parameter analysis is applicable when internal conduction resistance within the solid is very small as compared to the external convection resistance of the medium.
 (B) In lumped parameter analysis, the thermal conductivity of the solid body is very high.
 (C) In lumped parameter analysis problems, the time- temperature variation curve is parabolic.
 (D) Heisler chart method is used when internal conduction resistance within the solid is very high as compared to the external convection resistance of the medium.
- Q7** A 2 m wide and 0.5 cm thick steel plate coming out from a furnace at 820°C is to be quenched in an oil bath ($c_p = 2 \text{ kJ/kg}^{\circ}\text{C}$) at 45°C . Plate is moving at a steady velocity of 15 m/min and the oil bath is 9 m long. Convection heat transfer coefficient between plate and oil is $860 \text{ W/m}^2\text{K}$. For Plate $k = 60.5 \text{ W/m}^{\circ}\text{C}$, $\rho = 7854 \text{ kg/m}^3$, $c_p = 434 \text{ J/kg}^{\circ}\text{C}$. The temperature of the plate when it leaves the oil bath (in $^{\circ}\text{C}$) is ____ (round off to one decimal place).
- Q8** A hot cylinder ingot of 50 mm diameter and 200 mm length is taken out from the furnace at 800°C and then dipped in water till its



temperature falls to 500°C ($h_w = 200 \text{ W/(m}^2 \text{ K)}$). Then it is directly exposed to air till its temperature falls to 100°C ($h_a = 20 \text{ W/m}^2 \text{ K}$). The temperature of air and water is 30°C . Taking the properties of ingot as $\rho = 800 \text{ kg/m}^3$,

$c = 0.2 \text{ kJ/kg-K}$, $k = 60 \text{ W/mK}$, the total time required for the ingot to reach the temperature from 800°C to 100°C is ___ min (round off to two decimal places).



Answer Key

Q1 (A, B, C, D)

Q2 (B)

Q3 0.7~0.85

Q4 16~17

Q5 (A)

Q6 (A, B)

Q7 65~66

Q8 2.75~3



[Android App](#) | [iOS App](#) | [PW Website](#)



ME & XE (Hinglish)

Heat Transfer

DPP: 2

Radiation Heat Transfer

Q1 A sphere (surface 1) of 2 m diameter at 240 K is kept in an evacuated cubic enclosure (surface 2) whose sides are 3 m long. The emissivity of the sphere & the enclosure is 0.1 & 0.8 respectively. If the temperature of the cubic enclosure is 100 K, the net rate of radiation heat transfer will be (in W) _____ (Correct up to two decimals). Take Stefan Boltzmann constant $5.67 \times 10^{-8} \text{ W/m}^2\text{K}^4$.

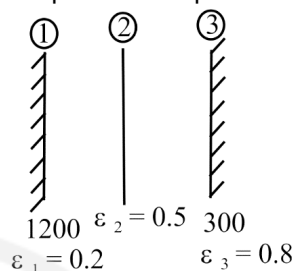
Q2 An enclosure of dimension 2 m $\times$ 3 m with a height of 2 m. The walls and ceilings are maintained at 250° and the floor is at 120°C . The walls and ceiling have an emissivity of 0.8 and the floor have 0.5. The net radiation heat transfer (in kW) is ____ (round off to one decimal place).

Q3 A black body of total area 0.045 m^2 is completely enclosed in a space bounded by 5 cm thick wall. The walls have a surface area 0.5 m^2 and thermal conductivity $1.07 \text{ W/m}^\circ\text{C}$. If the inner surface of the enveloping wall is to be maintained at 215°C and the outer wall surface is at 30°C . Neglect the difference between inner and outer surface area of the enveloping wall. The temperature of the black body is (in $^\circ\text{C}$) _____ (round off to nearest integer).

Q4 The floor of a hemispherical furnace (radius 1 m) is at 800 K and has an emissivity of 0.5. The corresponding values for the hemispherical roof are 1200 K and 0.25. The net radiation heat transfer from the roof to floor is (in kW) _____ (round off to the nearest integer).

Q5 Three infinitely long parallel plates are arranged as shown in the figure. The plate 1 and 3 are maintained at 1200 K and 300 K respectively. The emissivities of the three plates are 0.2, 0.5

and 0.8 respectively. If the plate 2 received no heat from external source, what will the temperature of plate 2?



- (A) 300 K (B) 900 K
(C) 869 K (D) 750 K

Q6 What is the basic equation of thermal radiation from which all other equations of radiations can be derived?

- (A) Stefan-Boltzmann equation
(B) Planck's equation
(C) Wein's equation
(D) Rayleigh-jeans equation

Q7 For an opaque surface, the irradiation, radiosity and emissive power are respectively 20, 12 and 10 W/m^2 . What is the emissivity of the surface?

- (A) 0.2 (B) 0.4
(C) 0.9 (D) 1.0

Q8 A thermocouple whose surface is diffuse and gray with an emissivity of 0.6 indicates a temperature of 180°C when used to measure the temperature of a gas flowing through a large duct whose walls have a uniform temperature of 450°C . If the Convection heat transfer coefficient between the thermocouple and the gas stream is $125 \text{ W/m}^2\text{K}$ and negligible conduction losses from the thermocouple, the temperature of the gas is _____ $^\circ\text{C}$ (round off to one decimal place).



Answer Key

Q1 225~230

Q2 8.2~8.7

Q3 670~690

Q4 83~87

Q5 (C)

Q6 (B)

Q7 (C)

Q8 115~120



[Android App](#)

| [iOS App](#)

| [PW Website](#)



ME & XE (Hinglish)

Heat Transfer

DPP: 1

Radiation Heat Transfer

- Q1** Choose the correct statement
- (A) Hollow sphere with uniform temperature and a small hole in it behaves as black body.
- (B) A body which partly absorb and partly reflect, but does not allow any radiation to pass through it is called Gray body.
- (C) When a body is in thermal equilibrium with its surroundings, the emissivity of its surface is equal to its absorptivity. This is called Planck's law.
- (D) Ice is very close to the black body.
- Q2** Choose the correct statement
- (A) The wavelength at which monochromatic emissive power of a black body is maximum shifts towards right with increasing temperature.
- (B) The maximum monochromatic emissive power of a black body is directly proportional to the T^4 .
- (C) A blackbody is perfect emitter.
- (D) Planck's equation is the basic equation of thermal radiation from which all other equations of radiation can be derived.
- Q3** A Gray body is the body whose emissivity
- (A) varies with temperature.
- (B) varies with the wavelength of incident ray.
- (C) varies with temperature and wavelength of incident rays.
- (D) does not vary with temperature and wavelength of incident rays.
- Q4** Radiation energy with an intensity of 800 W/m^2 strikes a flat plate. The absorptivity is twice the transmissivity and thrice the reflectivity. Rate of absorption of energy is (in W/m^2) _____ (round off to one decimal place).
- Q5** A radiation shield is placed between two infinite parallel plates of temperature T_1 and T_2 . Emissivities of radiation shield and plate are same. The ratio of radiation energy flux with shield and without shield would be
- (A) 0.25 (B) 0.5
- (C) 0.75 (D) 1
- Q6** A thermos flask has a double walled bottle and the space between the walls is evacuated so as to reduce the heat flow. The bottle surfaces are silver plated and the emissivity of each surface is 0.025. If the content of the bottle is at 375 K, the rate of heat loss from the thermos flask to the ambient air at 300 K (in W/m^2) is _____ (round off to two decimal places).
- Q7** Consider radiative heat transfer between two large parallel plates of surface emissivity 0.8. How many thin radiation shields of emissivity 0.05 be placed between the surfaces to reduce the radiation heat transfer by a factor of 75?
- (A) 2 (B) 3
- (C) 4 (D) 5
- Q8** Consider a cylindrical enclosure whose height is twice the diameter of its base. If the view factor from its base to its top surface is 0.05, the view factor from the side surface of this cylindrical enclosure to its base surface will be
- (A) 0.95 (B) 0.475
- (C) 0.119 (D) 0.238



Answer Key

Q1 (A, D)

Q2 (C, D)

Q3 (D)

Q4 435~438

Q5 (B)

Q6 8~8.3

Q7 (B)

Q8 (C)



[Android App](#)



[iOS App](#)



[PW Website](#)



ME & XE (Hinglish)

Heat Transfer

DPP: 2

Convection Heat Transfer

Q1 For laminar flow of a fluid along a flat plate, one would expect the largest local convection heat transfer coefficient for the same Reynolds and Prandtl number when

- (A) Constant temperature is maintained on the surface
- (B) Constant heat flux is maintained on the surface
- (C) Plate surface is polished
- (D) None of these.

Q2 Which of the following dimensionless parameter pair is significant for forced convection and natural convection, respectively?

- (A) Reynolds, Grashof
- (B) Reynolds, Mach
- (C) Reynolds, Eckert
- (D) Reynolds, Schmidt

Q3 For flow of air at 15°C and 1 atm over a 0.3 m wide plate at 65°C with a velocity of 3m/s, match the following quantities (List I) with corresponding computed values (List II) at $x = 0.3$ m. Take air properties as density $= 1.127 \text{ kg/m}^3$, dynamic viscosity $= 1.918 \times 10^{-5} \text{ kg/m-s}$ and $Pr = 0.7255$

List I	List II
A. Hydrodynamic boundary layer thickness (in m)	1. 0.0058
B. Average friction coefficient	2. 0.0065
C. Average Shear stress due to friction (in N/m^2)	3. 0.00726
D. Thermal boundary layer thickness (in m)	4. 0.0294

Code

A	B	C	D
(A) 4	1	2	3
(B) 2	3	4	1
(C) 4	3	2	1
(D) 2	1	4	3

Q4 The convective heat transfer coefficient for laminar & turbulent flows over a flat plate varies with x respectively as (where x is the distance from leading edge of the plate)

- (A) $x^{1/2}$ & $x^{0.2}$
- (B) $x^{1/2}$ & $x^{-0.2}$
- (C) $x^{-1/2}$ & $x^{-0.2}$
- (D) $x^{1/3}$ & $x^{0.2}$

Q5 For liquid Ammonia at 20°C the thermo-physical properties are: density $= 611.75 \text{ kg/m}^3$, kinematic viscosity $= 0.359 \times 10^{-6} \text{ m}^2/\text{s}$, specific heat $4.798 \text{ kJ/kg}^\circ\text{C}$ and thermal conductivity $0.521 \text{ W/m}^\circ\text{C}$. Ratio of thermal boundary layer thickness to hydrodynamic boundary layer thickness is _____. (Assume Laminar condition)

Q6 An object ($T_s = 400 \text{ K}$) of irregular shape has characteristic length of 1m. When this object is placed in atmospheric air ($T_\infty = 300 \text{ K}$) which is moving with a velocity of 100 m/s, average convective heat flux is $20,000 \text{ W/m}^2$. If the second object ($T_s = 400 \text{ K}$) of same shape but with a characteristic length of 5m is placed in atmosphere (300 K) which is moving with a velocity of 20 m/s then the average convective coefficient in case of second object is ____ $\text{W/m}^2\text{K}$ (round off to nearest integer).

Q7 A thin flat plate that is $0.2\text{m} \times 0.2\text{m}$ on a side is oriented parallel to an atmospheric air stream having a velocity of 40 m/s. The air is at a



temperature of 20°C , while the plate is maintained at 120°C . The air flows over the top and bottom surfaces of the plate, the measurement of the drag force shows a value of 0.075 N . The rate of heat transfer from both sides of the plate to the air is ____W. Take properties of air at 70°C , $\rho = 1.018\text{ kg/m}^3$, $c_p = 1009\text{ J/kg-K}$, $\text{Pr} = 0.70$.

- Q8** A Steel Plate of thickness 6mm and cross section of $1\text{m} \times 1\text{m}$ is heat treated by cooling it with

atmospheric air of velocity $U_{\infty} = 10\text{ m/s}$ and temperature $T_{\infty} = 20^{\circ}\text{C}$ in parallel flow over a plate Initial temperature of the plate is 300°C . Properties of steel Plate are, $k = 49.2\text{ W/(m-K)}$, $c = 549\text{ J/kg-K}$, $\rho = 7832\text{ kg/m}^3$ and for air properties are $\nu = 30.4 \times 10^{-6}\text{ m}^2/\text{s}$, $k = 0.036\text{ W/m-K}$, $\text{Pr} = 0.688$.

The magnitude of rate of change of temperature at the beginning is ____K/s (round off to two decimal places).



Answer Key

Q1 (B)

Q2 (A)

Q3 (D)

Q4 (C)

Q5 7.7~8.1

Q6 38~42

Q7 234~239

Q8 0.24~0.28



[Android App](#)



[iOS App](#)



[PW Website](#)



ME & XE (Hinglish)

Heat Transfer

DPP: 1

Convection Heat Transfer

Q1 Consider the flow of fluid over a flat plate maintained at constant temperature. If the free stream velocity of the fluid is doubled then what will be the ratio of the rate of heat transfer between the fluid and the plate? (Assume laminar condition)

- (A) 2 (B) 1.414
(C) 2.5 (D) 0.5

Q2 Choose the correct statement(s).

(A) In convection heat transfer, the surface heat transfer coefficient from a heated surface is a function of Re, Gr and Pr.

(B) For combined forced and free convection, forced convection is predominant if $\frac{Gr}{Pr^2} \gg 1$

(C) For combined forced and free convection, forced convection is Gr predominant if $\frac{Gr}{Pr^2} \ll 1$

(D) In free convection, the surface heat transfer coefficient from a heated surface is a function of Re, Gr.

Q3 Air at 300 K and 1 atm flows over a flat plate maintained at 426 K at a velocity of 30 m/s. If air has $\mu = 2.13 \times 10^{-5}$ kg/m-s, $Pr = 0.7$, how thick will be the thermal boundary layer at a distance of 2.5 cm from the leading edge of the plate?

- (A) 0.470 mm (B) 0.620 mm
(C) 0.760 mm (D) 0.892 mm

Q4 A fluid is flowing along a plate which is maintained at constant temperature and flow is laminar. The heat transfer coefficient along the length

- (A) Decreases
(B) Increases

(C) Remain constant

(D) First decrease and then increase

Q5 Air at a temperature of T_∞ flow over a flat plate with a free stream velocity of U_∞ . The plate is maintained at a constant temperature of T_w . The velocity (u) & temperature (T) of air at any location are given by

$$\frac{u}{U_\infty} = \frac{y}{\delta} \quad \& \quad \frac{T - T_w}{T_\infty - T_w} = 2 \left(\frac{y}{\delta_t} \right) - \left(\frac{y}{\delta_t} \right)^2$$

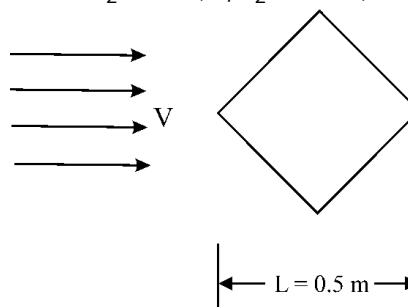
Where y is the distance measured from the plate along its normal, and δ & δ_t are the hydrodynamic & thermal boundary layer thickness respectively. Ratio of heat transfer coefficient to shear stress at the plate surface is given by $\frac{Ck}{\mu U_\infty} Pr^{\frac{1}{3}}$ value of C is

- (A) 4 (B) 1/4
(C) 2 (D) 1/2

Q6 Experimental measurements of the convection heat transfer coefficient for a square bar of characteristic length 'L' in cross flow give the following values.

When $V_1 = 20$ m/s, $h_1 = 50$ W/m²K

When $V_2 = 15$ m/s, $h_2 = 40$ W/m²K



Assume that the functional form of the Nusselt number is $Nu = CRe^a Pr^b$. Convection heat transfer coefficient for a similar bar with $L = 1$ m and at $V = 15$ m/s is _____ W/m²K (round off to two decimal places).



Q7 Air at a free stream temperature of $T_\infty = 20^\circ\text{C}$ is in parallel flow over a flat plate of length $L = 5\text{ m}$ and temperature $T_s = 90^\circ\text{C}$. Spatial variation of temperatures measured in the boundary layer is expressed as $T(^{\circ}\text{C}) = 20 + 70 \exp(-600 xy)$ where x is distance from the leading edge of plate (in m) and y is distance normal to the plate (in m). Average convective heat transfer coefficient for the plate is _____ $\text{W/m}^2\text{K}$ (round off to nearest integer). For air $k = 0.03\text{ W/mk}$.

Q8 For a parallel flow of atmospheric air over a flat plate of length , the local convective heat transfer coefficient may be expressed as $h_x = 0.7 + 13.6x - 3.4 x^2$, where h_x has unit of $\text{W/m}^2\text{K}$ and is the distance from leading edge of the plate (in meter). Ratio of the average convection coefficient up to length L to maximum convection coefficient is _____ (round off to two decimal places).



Answer Key

Q1 (B)

Q2 (A, C)

Q3 (C)

Q4 (A)

Q5 (C)

Q6 34~34.5

Q7 44~46

Q8 0.7~0.8



[Android App](#) | [iOS App](#) | [PW Website](#)



ME & XE (Hinglish)

Heat Transfer

DPP: 1

Heat Exchanger

- Q1** For evaporator and condenser type of heat exchanger under given condition, LMTD for counter flow will be
 (A) Greater than parallel flow
 (B) Less than parallel flow
 (C) Equal to parallel flow
 (D) Very much larger than parallel flow
- Q2** In a concentric double-pipe heat exchanger where one of the fluids undergoes phase change
 (A) the two fluids should flow opposite to each other.
 (B) the two fluids should flow parallel to each other.
 (C) the two fluids should flow normal to each other.
 (D) the direction of flow of the two fluids are of no consequence.
- Q3** Two fluid A and B exchange heat in a counter flow heat exchanger. Fluid A enters at 420°C and has a mass flow rate of 1 kg/s . Fluid B enters at 20°C and has a mass flow rate of 1 kg/s . The effectiveness of heat exchanger is 75% . Specific heat of fluid A is 1 kJ/kgK and that of fluid B is 4 kJ/kgK . Exit temperature of fluid B (in $^{\circ}\text{C}$) is ____ (round off to nearest integer).
- Q4** In a solar assisted air-conditioning system, 0.5 kg/s of ambient air at 270 K is to be preheated by the same amount of air leaving the system at 295 K . If a counter flow heat exchanger has an area of 30 m^2 and the overall heat transfer coefficient is $25\text{ W/m}^2\text{K}$. Take c_p for air as 1000 J/kg K . Outlet temperature of the preheated air (in K) is ____ (round off to nearest integer).
- Q5** A chemical having specific heat of 3.3 kJ/kgK flowing at a rate of $20,000\text{ kg/h}$ enters a parallel flow heat exchanger at 120°C . The flow rate of cooling water is 50000 kg/h with an inlet temperature of 20°C . The heat transfer area is 10 m^2 and the overall heat transfer coefficient is $1050\text{ W/m}^2\text{K}$. The outlet temperature of chemical is (in $^{\circ}\text{C}$) ____ (round off to nearest integer).
- Q6** A counterflow heat exchanger is to heat air entering at 400°C with a flow rate of 6 kg/s by the exhaust gas entering at 800°C with a flow rate of 4 kg/s . The overall heat transfer coefficient is $100\text{ W/m}^2\text{K}$ and the outlet temperature of air is 550°C . Specific heats for both air and exhaust gas can be taken as $1100\text{ J/kg}\cdot\text{K}$. The Number of transfer Units is ____ (round off to two decimal places).
- Q7** A flow of 0.1 kg/s of exhaust gases at 700 K from a gas turbine is used to preheat the incoming air, which is at the ambient temperature of 300 K . It is desired to cool the exhaust to 400 K , and it is estimated that an overall heat transfer coefficient of $30\text{ W/m}^2\text{K}$ can be achieved in an appropriate exchanger. The area required for a counterflow heat exchanger is ____ m^2 (round off to nearest integer). Take the specific heat of exhaust gases the same as for air, which is 1000 J/kgK .
- Q8** A coaxial tube counterflow heat exchanger is to cool 0.03 kg/s of benzene from 360 K to 310 K with a counterflow of 0.02 kg/s of water at 290 K . If the inner tube outside diameter is 2 cm and the overall heat transfer coefficient based on outside area is $650\text{ W/m}^2\text{K}$, the required length of the exchanger is ____ m (round off to one decimal place). Take the specific heats of benzene and water as 1880 and 4175 J/kgK , respectively.



Answer Key

Q1 (C)
Q2 (D)
Q3 93~97
Q4 280~290

Q5 78~82
Q6 1~1.2
Q7 9~11
Q8 2.4~2.7



[Android App](#) | [iOS App](#) | [PW Website](#)

